Section 4.3

4.3–1

Joint pmf:

$$f(x,y) = \frac{x+y}{32}, \qquad x = 1, 2,\ y = 1, 2, 3, 4.$$

(a) Joint table and marginals

$f(x,y)$	$y=1$	$y=2$	$y=3$	$y=4$
$x=1$	$\frac{1}{16}$	$\frac{3}{32}$	$\frac{1}{8}$	$\frac{5}{32}$
$x=2$	$\frac{3}{32}$	$\frac{4}{32}$	$\frac{5}{32}$	$\frac{3}{16}$

$$P(X=1) = \frac{14}{32} = \frac{7}{16}, \qquad P(X=2) = \frac{18}{32} = \frac{9}{16}.$$

$$p_Y(y) = \frac{3+2y}{32}, \quad y = 1, \ldots, 4.$$

$$p_Y(1) = \tfrac{5}{32},\ p_Y(2) = \tfrac{7}{32},\ p_Y(3) = \tfrac{9}{32},\ p_Y(4) = \tfrac{11}{32}.$$

(b) Conditional pmf of $X|Y=y$:

$$g(x \mid y) = \frac{f(x,y)}{p_Y(y)} = \frac{x+y}{3+2y},$$

$$g(1 \mid y) = \frac{1+y}{3+2y}, \qquad g(2 \mid y) = \frac{2+y}{3+2y}.$$

(c) Conditional pmf of $Y|X=x$:

For $x = 1$:

$$h(y \mid 1) = \frac{(1+y)/32}{7/16} = \frac{1+y}{14}.$$

For $x = 2$:

$$h(y \mid 2) = \frac{(2+y)/32}{9/16} = \frac{2+y}{18}.$$

(d) Requested probabilities

$$P(1 \le Y \le 3 \mid X = 1) = \sum_{y=1}^{3} \frac{1+y}{14} = \frac{9}{14}.$$

$$P(Y \le 2 \mid X = 2) = \frac{3}{18} + \frac{4}{18} = \frac{7}{18}.$$

$$P(X = 2 \mid Y = 3) = \frac{5}{9}.$$

(e) $E[Y \mid X = 1]$ **and** $\mathrm{Var}(Y \mid X = 1)$

$$E[Y \mid 1] = \frac{\sum_{y=1}^{4} y(1+y)}{14} = \frac{40}{14} = \frac{20}{7}.$$

$$E[Y^2 \mid 1] = \frac{130}{14} = \frac{65}{7}.$$

$$\mathrm{Var}(Y \mid 1) = \frac{65}{7} - \left(\frac{20}{7}\right)^2 = \frac{55}{49}.$$

4.3–2

Joint pmf nonzero points:

$$(1,1) = \tfrac{3}{8},\ (2,1) = \tfrac{1}{8},\ (1,2) = \tfrac{1}{8},\ (2,2) = \tfrac{3}{8}.$$

$$P(X = 1) = P(X = 2) = P(Y = 1) = P(Y = 2) = \tfrac{1}{2}.$$

Conditional pmfs of $X \mid Y$**:**

$$P(X = 1 \mid Y = 1) = \tfrac{3}{4}, \quad P(X = 2 \mid Y = 1) = \tfrac{1}{4}.$$

$$P(X = 1 \mid Y = 2) = \tfrac{1}{4}, \quad P(X = 2 \mid Y = 2) = \tfrac{3}{4}.$$

Means and variances:
For $Y = 1$:

$$E[X \mid 1] = \tfrac{5}{4}, \qquad E[X^2 \mid 1] = \tfrac{7}{4}, \qquad \mathrm{Var}(X \mid 1) = \tfrac{3}{16}.$$

For $Y = 2$:

$$E[X \mid 2] = \tfrac{7}{4}, \qquad E[X^2 \mid 2] = \tfrac{13}{4}, \qquad \mathrm{Var}(X \mid 2) = \tfrac{3}{16}.$$

Conditional pmfs of $Y \mid X$:

$$P(Y=1 \mid X=1)=\tfrac{3}{4}, \quad P(Y=2 \mid X=1)=\tfrac{1}{4},$$
$$P(Y=1 \mid X=2)=\tfrac{1}{4}, \quad P(Y=2 \mid X=2)=\tfrac{3}{4}.$$

$$E[Y \mid X=1]=\tfrac{5}{4}, \quad E[Y \mid X=2]=\tfrac{7}{4}, \quad \mathrm{Var}(Y \mid X)=\tfrac{3}{16}.$$

Section 5.1

5.1–6

PDF: $f_X(x)=2x,\ 0<x<1.$

(a) CDF

$$F_X(x)=\int_0^x 2t\,dt = x^2.$$

(b) Inverse transform simulation

$$X=\sqrt{U}, \qquad U \sim \mathrm{Unif}(0,1).$$

(c) Example simulation

$$U=\{0.01, 0.04, 0.09, 0.16, 0.25, 0.36, 0.49, 0.64, 0.81, 0.99\}.$$

$$X=\{0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 0.99499\}.$$

5.1–7

Given X has pdf

$$f_X(x)=\theta x^{\theta-1}, \quad 0<x<1.$$

Let $Y=-2\theta \ln X$. Then:

$$x=e^{-y/(2\theta)}, \qquad \frac{dx}{dy}=-\frac{1}{2\theta}e^{-y/(2\theta)}.$$

$$f_Y(y)=\frac{1}{2}e^{-y/2}, \qquad y>0.$$

Thus $Y \sim \mathrm{Exp}(\lambda=\tfrac{1}{2})$.

5.1–12

$X \sim U(-1,3) \Rightarrow f_X(x) = 1/4$ on $[-1,3]$. Let $Y = X^2$, so $0 \le y \le 9$.

For $0 \le y \le 1$, both roots are in support, so

$$f_Y(y) = \frac{1}{4}\left(\frac{1}{2\sqrt{y}} + \frac{1}{2\sqrt{y}}\right) = \frac{1}{4\sqrt{y}}.$$

For $1 < y \le 9$, only $+\sqrt{y}$ is in support:

$$f_Y(y) = \frac{1}{8\sqrt{y}}.$$

Elsewhere, $f_Y(y) = 0$.